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OPEN TENDER PRODUCT SUGGESTION SYSTEM -- FLOW

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STEP 1 -- ZIP UPLOAD

Intern receives the tender ZIP file from the open tender portal and uploads it to the system.

Input : Tender ZIP file
(contains bid document PDF + ATC files)

STEP 2 -- ZIP EXTRACT & DOCUMENT CATEGORIZE

System extracts the ZIP and sorts the files inside:

Bid Document (PDF) -> **Main tender details**
Item list, quantities, buyer info

ATC Files (PDF / DOCX / XLSX)
-> **Additional Terms & Conditions**
added by the buyer
** PRIMARY source of technical specs

Catalogue Links (URL inside PDF)
-> **GEM catalogue page with spec table**
(scraped automatically)

STEP 3 -- FIND NUMBER OF PRODUCTS

System reads the bid document / ATC / CSV (BOQ) to count how many individual product items are in the tender.

If a CSV / BOQ file is present -> item count comes from CSV
If only PDF / DOCX -> item count from bid text

Output : Total number of relevant product items found
e.g. "3 items found"

STEP 4 -- EACH PRODUCT : DEPT CATEGORIZE

For each product item found, the system classifies it into the correct Meril department and category:

DEPT : ENDO

Types : Suture, Stapler, Mesh, IUDs, Energy Device,
Biosurgicals, Glue, Clutch, Kit, Others

DEPT : DIAGNO

Analyzer : 3-Part / 5-Part Hematology Analyzer,

Biochemistry Analyzer, ELISA Plate Washer,
Electrolyte Analyzer, HPLC HbA1c Analyzer,
Fluorescence Immunoassay Analyzer, e-CLIA

Reagents : **Hematology Reagents, CelQuant Series,**
ClotQuant, FloQuant, GluQuant, QuantiLyte,
TIA, SAA, FSR, PRV, ER, FDR, BS120/200

RapidELISA: ELISA Kits, Rapid Test Kits

System : **AQ 100/200 System Packs,**
Packs **AQ 400 System Packs**

-> If item does NOT match any Meril category
Result = "Not Relevant" -- no further processing

STEP 5 -- PRODUCT SUGGESTION

For each categorized item:

5a. PICK PRODUCT FILE

System selects the correct product catalog JSON

based on dept + type

**e.g. Suture -> products/endo/sutures_by_material/
3-Part Analyzer -> products/analyzer/3-Part...json**

5b. MATCH BEST PRODUCT

AI (GPT / Ollama) reads the product catalog and picks
the SINGLE closest matching Meril product based on:

- Material, Size, Type
- Needle type / shape (for sutures)
- Performance specs (for analyzers)

5c. DEVIATION ANALYSIS

AI compares EVERY tender technical specification
against the matched product spec:

Complied -> **Product meets the requirement**
Deviation -> **Product does not meet it (gap noted)**
Not Specified -> **Spec not in Meril product database**

Output per item:

- Product Code
- Product Name
- Relevancy Score (0.0 to 1.0)
- Deviation Table (spec by spec comparison)

STEP 6 -- SAVE RESULTS

All results saved to database:

Table : tender_processing_results

Fields : Bid No, Result (yes/no), Dept, Category,
Evaluation JSON, Suggested Products,
Deviation Tables

Also saved as JSON files in OUTPUT/ folder:

<bid>_evaluation.json

<bid>_product_suggestions.json

<bid>_deviation_analysis.json

QUICK SUMMARY

ZIP Upload

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Extract + Categorize Documents

(Bid PDF | ATC Files | Catalogue Links | CSV BOQ)

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Count Products in Tender

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v

For Each Product -- Classify Dept + Type

(Endo / Diagno -- or mark Not Relevant)

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Match Best Meril Product from Catalog

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v

Run Deviation Analysis (Complied / Deviation / Not Specified)

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Save to Database + Output JSON Files
